Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · HAUSMAN TEST

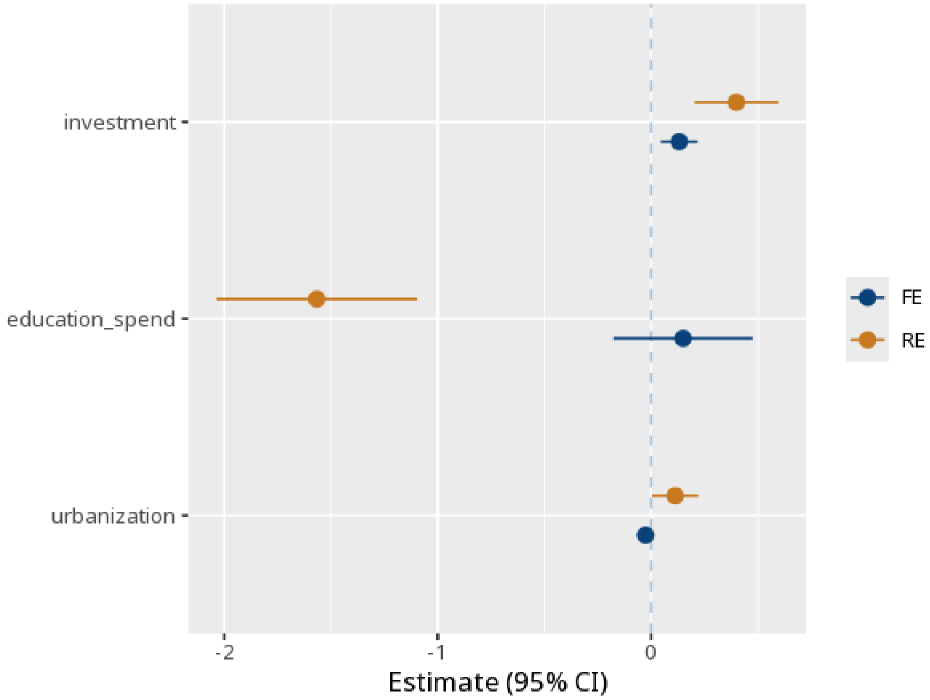
fixed vs. random effects?

Table 1. Hausman test

	Fixed effects	Random effects	Difference
investment	0.13	0.40	−0.27
	(0.04)	(0.10)	
	[0.04, 0.22]	[0.20, 0.59]	
education_spend	0.15	−1.57	1.72
	(0.16)	(0.24)	
	[−0.18, 0.47]	[−2.04, −1.10]	
urbanization	−0.03	0.11	−0.14
	(0.02)	(0.05)	
	[−0.06, 0.01]	[0.00, 0.22]	
Num.Obs.	100		
N entities	10		
R^2	.44	.93	

Hausman $\chi^2(3) = 572.56$, $p < .001$

Figure. FE vs. RE



Understanding the numbers

Fixed effects. The fixed-effects estimate for this predictor - consistent even if the random-effects assumption fails. Here, the fixed-effects estimate is 0.13.

Random effects. The random-effects estimate for the same predictor - more efficient, but only valid if entity effects are uncorrelated with the predictors. Here, the random-effects estimate is 0.40.

Difference. The difference between the fixed- and random-effects estimates - the quantity the Hausman test evaluates. Here, the difference is −0.27.

N. The total number of entity-period observations underlying both models. Here, N = 100.

N entities. The number of distinct entities in the panel. Here, N entities = 10.

R^2. Each model's own R^2, reported alongside the Hausman comparison. Here, R^2 is FE .44 / RE .93.

Hausman χ^2. The Hausman test statistic - how far the fixed- and random-effects estimates diverge, in aggregate across predictors. Here, $\chi^2(3) = 572.56$, $p < .001$.

How to read this test

Tests whether the random-effects assumption holds. A p below alpha means the estimates differ systematically → use fixed effects; a non-significant p means random effects is acceptable (and more efficient). Method: R plm package (Croissant & Millo, 2008). The standard errors in parentheses are clustered by entity; the bracketed line is the 95% CI. Your significance threshold (α) is 0.05.

APA template: A Hausman test comparing the fixed- and random-effects estimates gave $\chi^2(3)=572.56$, $p < .001$.

Statistical basis: Hausman specification test (Hausman, J. A. (1978). "Specification tests in econometrics." Econometrica, 46(6), 1251-1271.); Implementation (plm package) (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.). Full references in the exported CITATIONS.txt.